



CLINICAL RESEARCH STUDY

A randomized trial of a primary care–based disease management program to improve cardiovascular risk factors and glycated hemoglobin levels in patients with diabetes

Russell L. Rothman, MD, MPP,^{a,b} Robb Malone, PharmD, CDE,^c
Betsy Bryant, PharmD, CDE,^c Ayumi K. Shintani, PhD, MPH,^{a,b} Britton Crigler,^c
Darren A. Dewalt, MD,^c Robert S. Dittus, MD, MPH,^{a,f} Morris Weinberger, PhD,^{d,e}
Michael P. Pignone, MD, MPH^c

^aFrom the Division of General Internal Medicine, Department of Medicine, Center for Health Services Research, and

^bDiabetes Research and Training Center, Vanderbilt University Medical Center, Nashville, Tennessee;

^cDepartments of Medicine and

^dHealth Policy and Administration, University of North Carolina, Chapel Hill;

^eCenter for Health Services Research in Primary Care, Durham VAMC, North Carolina; and

^fVA Tennessee Valley Geriatric Research, Education and Clinical Center, Nashville.

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PURPOSE: To assess the efficacy of a pharmacist-led, primary care–based, disease management program to improve cardiovascular risk factors and glycated hemoglobin (A_{1C}) levels in vulnerable patients with poorly controlled diabetes.

METHODS: A randomized controlled trial of 217 patients with type 2 diabetes and poor glycemic control (A_{1C} level $\geq 8.0\%$) was conducted at an academic general medicine practice from February 2001 to April 2003. Intervention patients received intensive management from clinical pharmacists, as well as from a diabetes care coordinator who provided diabetes education, applied algorithms for managing glucose control and decreasing cardiovascular risk factors, and addressed barriers to care. Control patients received a one-time management session from a pharmacist followed by usual care from their primary care provider. Outcomes were recorded at baseline and at 6 and 12 months. Primary outcomes included blood pressure, A_{1C} level, cholesterol level, and aspirin use. Secondary outcomes included diabetes knowledge, satisfaction, use of clinical services, and adverse events.

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Requests for reprints should be addressed to Russell L. Rothman, MD, MPP, Internal Medicine and Pediatrics, Center for Health Services Research, Vanderbilt University Medical Center, Suite 6000, Medical Center East, Nashville, Tennessee 37232-8300.

E-mail address: russell.rothman@vanderbilt.edu.

RESULTS: For the 194 patients (89%) with 12-month data, the intervention group had significantly greater improvement than did the control group for systolic blood pressure (-9 mm Hg; 95% confidence interval [CI]: -16 to -3 mm Hg) and A_{1C} level (-0.8% ; 95% CI: -1.7% to 0%). Change in total cholesterol level was not significant. At 12 months, aspirin use was 91% in the intervention group versus 58% among controls ($P < 0.0001$). Intervention patients had greater improvements in diabetes knowledge and satisfaction than did control patients. There were no significant differences in use of clinical services or adverse events.

CONCLUSION: Our comprehensive disease management program reduced cardiovascular risk factors and A_{1C} levels among vulnerable patients with type 2 diabetes and poor glycemic control.

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Large studies such as the Diabetes Control and Complications Trial and the United Kingdom Prospective Diabetes Study (UKPDS) have demonstrated that intensive glucose control reduces the microvascular complications associated with diabetes.^{1,2} Other trials have also established the importance of addressing cardiovascular risk factors in patients with diabetes. Such risk reduction is critical because more than two thirds of patients with diabetes will die of cardiovascular causes and have 2 to 4 times the risk of coronary heart disease and stroke than persons without diabetes.^{3,4} Reducing cardiovascular risk for vulnerable patients, such as those with low socioeconomic status, low social support, or comorbid conditions, is particularly important, because these patients are at higher risk of diabetes-related complications and morbidity.^{4–7} Cardiovascular risk factor reduction, through intensive blood pressure control, cholesterol reduction, and aspirin use, can reduce the risk of cardiovascular events in patients with diabetes by as much as 50%.^{8–17}

Translating this evidence into practice can be difficult, however, and many patients receive inadequate care.^{18,19} One strategy involves disease management programs. Rather than the traditional model of health care delivery that often focuses on acute problems and visit-based care, disease management creates an “organized system tailored to the complex problems of chronic illness”.²⁰ Such programs are characterized by the use of multidisciplinary teams that provide integrated approaches to care, evidence-based care algorithms, and information systems that allow frequent tracking of patient-oriented outcomes and the adjustment of treatments.^{20–24} Most previous diabetes management programs have focused on glycemic control and have reduced glycated hemoglobin (A_{1C}) levels by as much as 1% to 2%.^{25–30} However, these programs should also address cardiovascular risk reduction if they are to be more beneficial. The primary care setting may be an advantageous site for multiple risk factor reduction because of the continuity of care provided and the emphasis on comprehensive management that includes medical and social issues.

We conducted a randomized controlled trial to assess the effects of a pharmacist-led, primary care–based, disease management program on cardiovascular risk factors and glucose control in vulnerable patients with poorly controlled diabetes.

Methods

Study design

This randomized trial was conducted at the University of North Carolina General Internal Medicine Practice, which serves a wide socioeconomic range of patients. The practice is staffed by more than 20 attending faculty and 70 medical residents who care for more than 2000 patients with diabetes. The study was initiated in February 2001 and was completed in April 2003. The institutional review board approved the study, and written informed consent was obtained from all participating patients.

Participants

Primary care providers at the practice referred eligible patients for possible participation. Patients were eligible if they were ≥ 18 years old, had a clinical diagnosis of type 2 diabetes, were followed for their diabetes care in the practice, had poor glucose control (A_{1C} level $\geq 8.0\%$), spoke English, and had a life expectancy > 6 months. We included patients with A_{1C} levels $\geq 8.0\%$ because we believed that these patients would most benefit from intensive glycemic control owing to their high risk of diabetes-related complications, including cardiovascular disease.^{31,32}

Procedures

Eligible patients participated in a 1-hour management session that was conducted by a clinical pharmacist practitioner from the disease management team. This session included diabetes education for the patient, and treatment recommendations were given to the patient's primary care provider. After this session, we randomly assigned patients to the intervention or control group using a random number generator; assignment was contained in sealed envelopes that were opened by the study coordinator. Patients assigned to the control group received usual care from their primary care provider and had no further management from the disease management team. In the intervention group, usual care was supplemented by intensive diabetes management (described below). Patients were followed for 1 year, with data collection occurring at baseline and 6 and 12 months.

Intervention

The intervention was delivered by 3 clinical pharmacist practitioners within the general medicine practice who had training in outpatient disease management; 2 were certified diabetes educators. The intervention included intensive educational sessions, evidence-based algorithms, and proactive management of clinical parameters. Clinical pharmacists provided patients with intensive education and counseling, as well as medication management. The results of these sessions were shared with the patient's primary care provider. Pharmacists were in contact with patients by telephone or in person every 2 to 4 weeks, or more frequently if indicated. They also had dedicated clinic slots to see patients directly or in consultation with an attending physician.

The pharmacists applied evidence-based treatment algorithms to help reduce cardiovascular risk factors and improve glycemic control. The algorithms were developed based on current medical evidence and input from the practice physicians. These algorithms, which were approved by the practice physicians, the hospital's pharmacy and therapeutics committee, and the institutional review board, allowed the pharmacists to initiate and increase use of blood pressure-, cholesterol-, and glucose-lowering medications (<http://www.med.unc.edu/medicine/edursrc/algor.htm>). All medication adjustments were done with the approval of the patient's primary care provider. Primary care providers could choose if they wanted to be contacted by telephone before making medication changes or if they wanted to receive written documentation after the changes.

A Microsoft Access (Redmond, Washington) database was developed to track patient outcomes and proactively improve care. For example, team members regularly queried the database to see which intervention patients had uncontrolled blood pressure, glucose, or cholesterol, thereby requiring more intensive intervention. The intervention group also had access to a diabetes care coordinator who was trained by the clinical pharmacists to address issues related to health behavior and health education. The coordinator called patients regularly to remind them of appointments and to assess if further interventions were indicated. The coordinator also helped to address barriers to care, including difficulties with transportation, communication, insurance problems, and low health literacy.

Measures

The primary outcomes of interest included blood pressure, A_{1C} levels and aspirin use at 6 and 12 months, and lipid levels at 12 months. Lipid and A_{1C} levels were obtained by the hospital laboratories, which were unaware of the study status of the patients. Laboratory values were calculated using a Boehringer Mannheim/Hitachi 911 automated discrete chemistry analyzer (Mannheim, Germany). For lipids, we measured total and high-density lipoprotein

cholesterol levels in all patients, and low-density lipoprotein (LDL) cholesterol levels only in patients who required statin therapy. Blood pressure was obtained by clinic nurses who were unaware of study assignment, using automated blood pressure machines (Welch Allyn, Beaverton, Oregon) and standard clinic practices. Aspirin use was obtained by self-report.

Secondary outcomes included diabetes knowledge, satisfaction, use of clinical services, and adverse events at 6 and 12 months, assessed through self-report. We measured diabetes knowledge with a scale that we developed and validated for patients with poor literacy. Satisfaction was measured using the Diabetes Treatment Satisfaction Questionnaire.³³ At enrollment, this questionnaire was not administered to patients with a recent (<3 months) diagnosis of diabetes. The use of medical services and adverse events were assessed by patient interview and supplemented by a review of the hospital clinical information system.

We used patient self-report to measure age, sex, race, insurance status, employment status, and the highest level of education completed. Hypertension was defined based on a diagnosis in the medical record (using *International Classification of Diseases, Ninth Revision* [ICD-9]) or if the patient met American Diabetes Association criteria (>130/85 mm Hg) at the time of the intervention. Hypercholesterolemia was also defined based on an ICD-9 diagnosis in the medical record or American Diabetes Association criteria (LDL cholesterol level >100 mg/dL).³⁴ Nephropathy was defined by chart documentation of nephropathy or spot urine microalbumin level >20 mg/dL. Retinopathy was defined from chart documentation or patient report. Neuropathy was defined by patient report, chart documentation, or a monofilament test with less than 10/10 correct with no callus.^{35,36}

To estimate resources required for our program, the study team also documented process measures for all intervention patients. This included recording time spent by the disease management team in direct contact with the patients, or in activities related to patient care (eg, scheduling appointments, reviewing charts) by the diabetes management team. The minimum time recorded for each activity was 5 minutes. The study team also documented all medication changes made in response to their management (eg, initiation of new medications, titration or alteration of current medications). These measures did not include any actions taken by the patients' primary care providers.

Sample size

Sample size estimates were based on a two-sided significance level of 0.05 and 80% power to detect between-group differences in the primary outcomes. Detection of a 10-mm Hg between-group difference in systolic blood pressure required a target sample of 93 patients per group. Detection of a 1% point difference in A_{1C} level or 20-mg/dL difference in total cholesterol level required 107 patients per group.

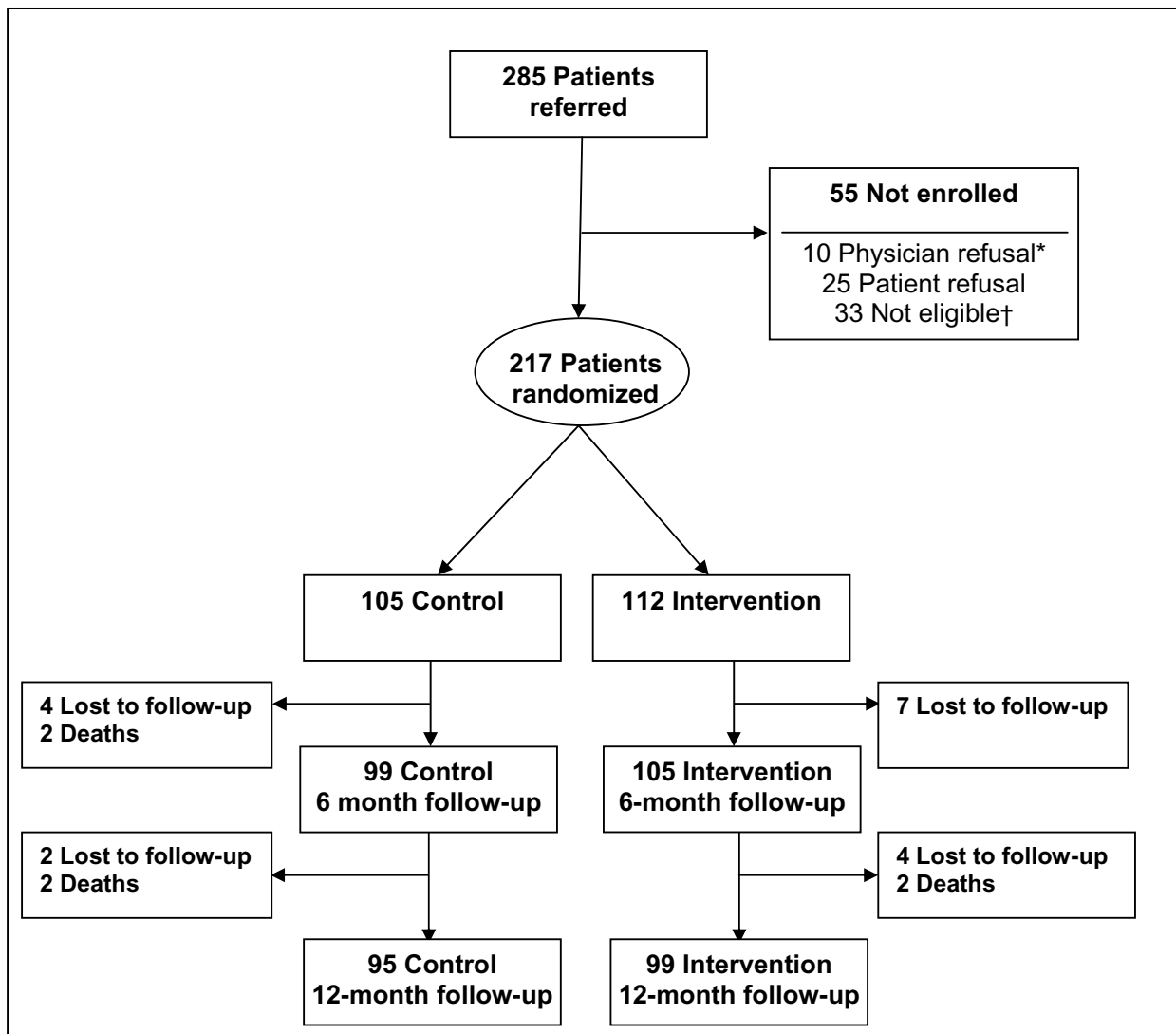


Figure 1 Study flow. The asterisk (*) indicates that the physician refused to sign consent and participate. The dagger (†) indicates that the patient was ineligible because of A_{1C} levels <8.0%, inability to speak English, or a life expectancy <6 months.

Statistical analysis

All analyses were performed using SAS 8.02 (SAS Institute, Cary, North Carolina), and a two-sided significance level of 5% was used for all statistical inference. All patients were analyzed based on their original assignments as an intention-to-treat analysis. Baseline differences between groups were assessed using *t* tests or Wilcoxon rank-sum tests for continuous outcomes and chi-squared tests for categorical outcomes.

For continuous outcomes, we performed repeated-measures analysis of variance³⁷ to account for data collection at baseline and 6 and 12 months. We also performed a 2-sample *t* test of differences (baseline to 12 months) between control and intervention groups. For all analyses, results were similar for both repeated-measures and 2-sample analyses; results of the 2-sample analyses are reported.

For cholesterol level, which was measured only at baseline and 12 months, 2-sample *t* tests and analysis of covari-

ance with adjustment for baseline measurement were performed. For aspirin use and statin use, we examined the proportion of eligible users at baseline and 12 months using chi-squared analyses. To evaluate the use of services and the frequency of adverse events, which were rare, negative binomial regression was performed.

Because of baseline differences after randomization, we further adjusted for age, race, and baseline values. Age and race were included because *P* values in the univariate analyses were less than 0.2.

Results

From February 2001 to April 2002, 217 patients were randomly assigned to the control or intervention group (Figure 1). Follow-up data were available for 204 patients (94%) at 6 months and 194 (89%) at 12 months, including 4 deaths in

Table 1 Baseline characteristics of the patients

Characteristic	Control (n = 105)	Intervention (n = 112)
	Number (%) or Mean $\pm$ SD	
Age (years)	57 $\pm$ 11	54 $\pm$ 13*
Women	59 (56)	63 (56)
African American	62 (59)	78 (70)†
Household income $\leq$ \$20 000	78 (74)	77 (69)
High school education or less	79 (75)	80 (71)
Attending physician as primary provider‡	37 (35)	33 (29)
Body mass index (kg/m ²)	34 $\pm$ 8	35 $\pm$ 9
Duration of diabetes (years)	9 $\pm$ 9	8 $\pm$ 9
Use of insulin at enrollment	40 (38)	45 (40)
Newly diagnosed diabetes ($\leq$ 3 months)	20 (19)	20 (18)
History of hypertension	87 (83)	93 (83)
History of hypercholesterolemia	66 (63)	67 (60)
History of nephropathy	50 (48)	45 (44)
History of retinopathy	27 (26)	26 (23)
History of neuropathy	43 (41)	41 (37)
Primary Measures		
Systolic blood pressure (mm Hg)	137 $\pm$ 21	140 $\pm$ 21
Diastolic blood pressure (mm Hg)	80 $\pm$ 11	82 $\pm$ 12
Glycated hemoglobin (%)	11 $\pm$ 3	11 $\pm$ 2
Total cholesterol (mg/dL)	201 $\pm$ 47	213 $\pm$ 84
Aspirin use	53 (50)	49 (44)
Statin use	44 (42)	35 (31)
Secondary Measures		
Diabetes knowledge (scale, 0 to 100)	49 $\pm$ 24	51 $\pm$ 24
Diabetes treatment satisfaction (scale, 0 to 36)	27 $\pm$ 5	29 $\pm$ 5
Weight (kg)	100 $\pm$ 24	101 $\pm$ 25
General visits§	2.4 $\pm$ 1.6	2.2 $\pm$ 1.5
Urgent care visits§	0.3 $\pm$ 0.7	0.4 $\pm$ 0.6
Emergency department visits§	0.4 $\pm$ 0.7	0.4 $\pm$ 0.7
Hospitalizations§	0.2 $\pm$ 0.6	0.3 $\pm$ 0.6
Hypoglycemic episodes§	0.4 $\pm$ 1.1	0.7 $\pm$ 2.0
Hypotensive episodes§	0.0 $\pm$ 0.3	0.1 $\pm$ 0.6

* $P = 0.05$; for all other comparisons $P > 0.20$.

† $P = 0.10$; for all other comparisons $P > 0.20$.

‡As opposed to having a resident physician supervised by an attending physician.

§Event rate for the 6 months before enrollment.

the control group and 2 deaths in the intervention group. Eighty-four physicians participated; no physician had more than 10 patients enrolled in the study.

Baseline characteristics were similar between the 2 groups and revealed a sample with poor glycemic control, multiple comorbid conditions, and low education and income status (Table 1). Clinical characteristics, including blood pressure, A_{1C} level, and aspirin use, were also similar. Approximately two thirds of the patients were African American, and over half were women. The intervention patients were slightly older than the control patients ($P = 0.05$) and more likely to be African American ($P = 0.10$).

Primary outcomes

Both systolic and diastolic blood pressure improved more among intervention than control patients (Figures 2A and B; Table 2). From baseline to 12-month follow-up, control patients had a 2-mm Hg increase in systolic blood pressure, while intervention patients had a 7-mm Hg decrease, a difference of 9 mm Hg (95% confidence interval [CI]: 3 to 16 mm Hg; $P = 0.008$). For diastolic blood pressure, control patients had an increase of 1 mm Hg, while intervention patients had a decrease of 4 mm Hg (difference, 5 mm Hg; 95% CI: 1 to 9 mm Hg; $P = 0.02$).

Glycated hemoglobin levels also improved more in the intervention group (Figure 2C; Table 2). From baseline to 12 months, control patients demonstrated a decrease of 1.6%, compared with a decrease of 2.5% among intervention patients (difference, 0.8%; 95% CI: 0% to 1.7%; $P = 0.05$).

Total cholesterol levels improved more in the intervention group, but the difference was modest and not statistically significant (Figure 2D; Table 2). At baseline, the rate of statin use was 42% (44/105) among control patients and 31% (35/112) among intervention patients; however, at 12-month follow-up, the rate was 47% (44/93) among control patients and 48% (47/99) among intervention patients ($P = 0.98$).

At 12-month follow-up, the use of aspirin for cardiovascular risk prevention was higher for intervention patients than control patients. Among control patients, aspirin was used in 58% (54/93) of eligible patients, whereas use was 91% (87/96) among intervention patients ($P < 0.0001$).

Secondary outcomes

Patients in the intervention group also had more improvements in diabetes knowledge and treatment satisfaction when compared with control patients (Table 2). There was no statistically significant difference in the use of services, such as general medicine visits, emergency department visits, or hospitalizations. There were also no statistically significant differences in potential adverse events, except for a slight increase in weight in the intervention group.

Process measures

The diabetes management team made a median of 45 contacts or care-related activities, a total of 460 minutes (38 minutes per month) for each intervention patient. This included time spent caring for patients and collecting data related to study evaluation. During this time, intervention patients had a median of three new medications added to their regimen by the disease management team and 4 titrations or adjustments to existing medications.

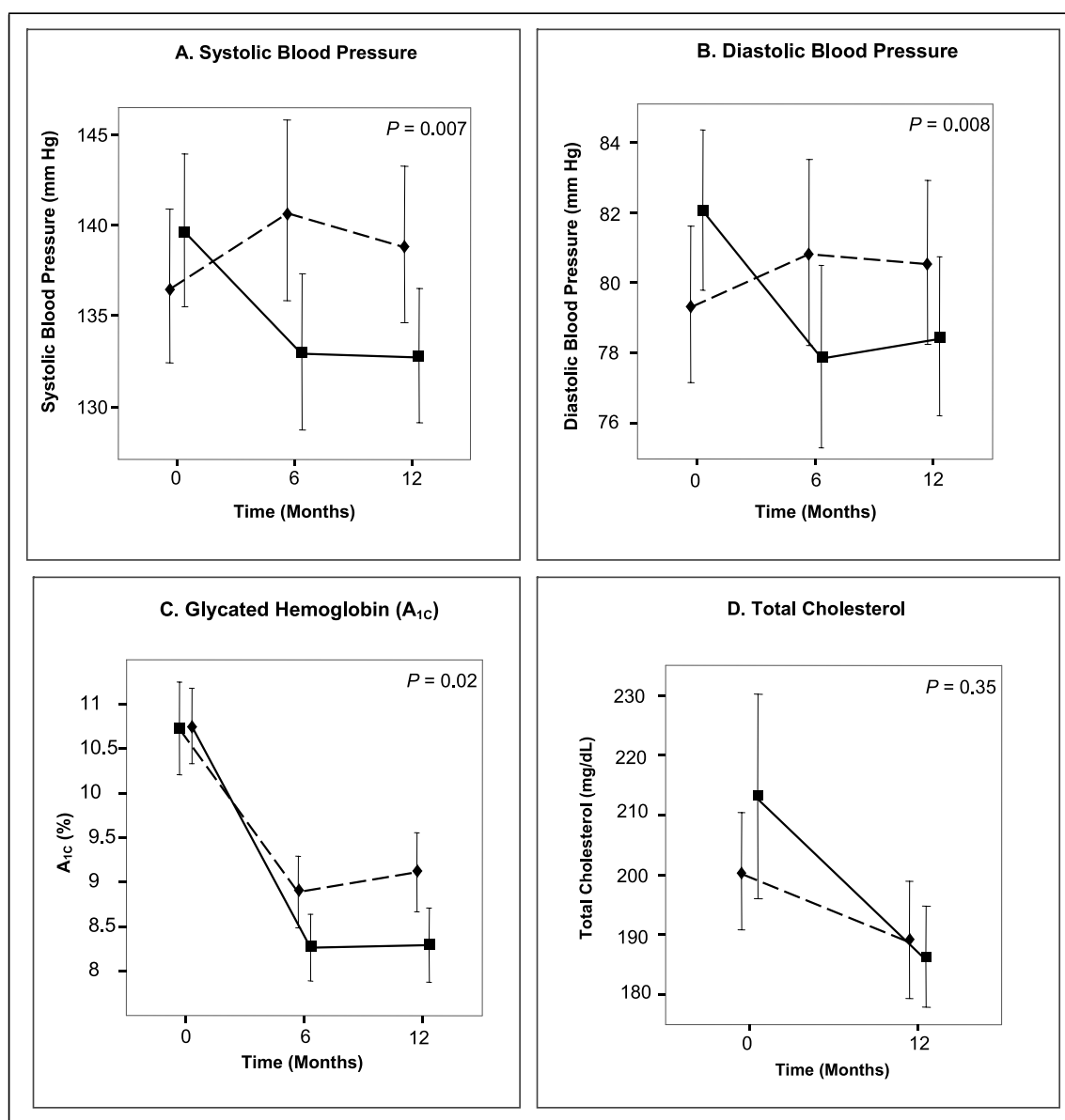


Figure 2 Change in clinical outcomes over time in the intervention (solid lines) and control (dashed lines) groups. A: systolic blood pressure (analysis of variance [ANOVA], $P = 0.007$); B: diastolic blood pressure (ANOVA, $P = 0.008$); C: glycated hemoglobin (A_{1c}) (ANOVA, $P = 0.02$); D: total cholesterol (analysis of covariance, $P = 0.35$). Mean values are presented.

Discussion

Our comprehensive pharmacist-led, primary care–based, disease management program improved blood pressure control, aspirin use, and glycemic control among vulnerable patients with type 2 diabetes. We observed a 9-mm Hg greater decrease in systolic blood pressure and a 5-mm Hg greater decrease in diastolic blood pressure in the intervention group than in the control group. In studies such as the Hypertension Optimal Treatment study and the UK-PDS,^{9–13} long-term decreases in blood pressure of this magnitude were associated with as much as a 50% reduction in cardiovascular events such as stroke or myocardial infarction.

The observed improvement in A_{1c} levels can reduce long-term microvascular complications among our patients. Although we did not observe a difference in cholesterol levels, there was a modest increase in statin use among the intervention patients. This lack of improvement in total cholesterol level may have been related to our algorithm for cholesterol management, which was developed in 2000 and consistent with the evidence at that time, but was probably not aggressive enough.

In terms of secondary outcomes, the intervention group had greater diabetes knowledge and treatment satisfaction. There were no differences in the utilization of clinical services or in adverse events, although the study was not powered to detect these differences. There was a

Table 2 Change in outcomes at 12 months

Variable	Control (n = 95)	Intervention (n = 99)	Difference or rate ratio (95% confidence interval)
Change in outcome from baseline to 12-month follow-up			
Systolic blood pressure (mm Hg)	+2	−7	−9 (−16 to −3)
Diastolic blood pressure (mm Hg)	+1	−4	−5 (−9 to −1)
Glycated hemoglobin (%)	−1.6	−2.5	−0.8 (−1.7 to 0)
Total cholesterol (mg/dL)	−12	−27	−15 (−35 to 4)
Diabetes knowledge	+13	+27	+14 (9 to 20)
Diabetes treatment satisfaction	+4	+8	+3 (1 to 6)
Weight (kg)	+0.1	+1.9	+1.8 (0.0 to 3.6)
Rate of event from 6- to 12-month follow-up			
General medicine visits	1.9	2.0	1.1 (0.9 to 1.3)
Urgent care visits	0.2	0.2	0.8 (0.4 to 1.6)
Emergency department visits	0.5	0.4	0.8 (0.5 to 1.4)
Hospitalizations	0.2	0.2	1.1 (0.6 to 2.0)
Hypoglycemic episodes	1.0	1.3	1.3 (0.6 to 2.5)
Hypotensive episodes	0.2	0.1	0.3 (0.1 to 1.6)

modest increase in weight, which may have been partly related to increased insulin use among the intervention patients.³⁸

We found that our diabetes management program improved cardiovascular risk factors in a predominantly minority sample of high-risk patients. Although a few diabetes specialty clinics have addressed the reduction of cardiovascular risk factors, most management programs have focused on glycemic control and have not targeted cardiovascular risk factors.^{25–30,39} This may be because diabetes is defined by abnormalities in glucose control, or because large randomized studies, which showed the benefit of glycemic control on microvascular risk reduction, have had a large influence on provider behavior.¹⁰ While intensive glycemic control is important for microvascular risk reduction, there have been only limited studies to suggest that intensive glucose control improves macrovascular outcomes.^{1,31} Conversely, intensive blood pressure control improves microvascular and macrovascular outcomes.^{3,9–12} This is particularly important because 80% of patients with type 2 diabetes will develop or die from macrovascular complications.⁹ One recent study estimated that intensive glycemic or cholesterol control has an incremental cost-effectiveness of \$40 000 to \$50 000 per quality-adjusted life-year (QALY), whereas intensive blood pressure control is not only cost-effective, but also saves almost \$2000 per QALY.⁴⁰

There are several limitations to our study. First, there were baseline differences between groups despite randomization. We tried to limit this concern by performing adjusted analyses, and these findings were similar to those from our unadjusted findings. Second, the study was conducted at a single academic medical center and follow-up was imperfect, both of which limit generalizability. Third, patients needed to have poor glycemic control at baseline to be eligible. Notably, improvements in both groups may in part be attributed to regression to the mean or the initial 1-hour management session. Despite these factors, we still

observed differences between the control and intervention groups. Finally, not all outcome assessment was blinded and several measures were based on patient self-report. We tried to limit these potential biases by focusing on objective measures for the clinical endpoints.

Our development of a new practice model, based fully in the primary care clinic, applies many of the recommendations for quality improvement put forth by the Institute of Medicine.⁴¹ The model improves quality of care by using information systems, evidence-based management, multidisciplinary teams, and a patient-centered approach that provides care both at and between clinic visits. Developing the model completely within a primary care clinic allows frequent communication among physicians, patients, and the diabetes management team. The model also targets patients with poor diabetes control and multiple barriers to care, a critical focus for primary care providers.

We believe that the success of our program, as compared with other recent programs,^{42,43} was due to several factors. The program allowed physicians to approve all anticipated medication changes at enrollment and to be notified after changes were made, an option that almost all physicians chose. This allowed mid-level providers to make frequent medication changes and overcome physician inertia.^{44,45} There was “face-to-face” time between patients and pharmacists (including at the initial visit), which allowed patients to build a trusting relationship with their mid-level provider, and this may have improved adherence. Finally, the use of clinical pharmacist practitioners may have led to better management of complex medication regimens. Although there is evidence that clinical pharmacists can be helpful for disease management programs for hypertension, hyperlipidemia, and other chronic conditions,^{46–49} evidence on the effectiveness of pharmacist care for diabetes is limited.^{45,50–53}

Our study demonstrates the effectiveness of a comprehensive, pharmacist-led, primary care–based disease management program for vulnerable patients with type 2 diabe-

tes. Unlike many previous diabetes management programs that focused primarily on glucose control, we addressed the most important sources of diabetes-related morbidity and mortality by targeting cardiovascular risk factors.

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